Name_____ Date_____ Period_____

**LESSON 5-5 · HOMEWORK PACKET · SHEET 1 OF 2**

# Volume of Rectangular Prisms

Maryland Standard 6.G.A.2 · Learning target – I can find the volume of a rectangular prism by multiplying length, width, and height.

**REMEMBER**

- **Volume** counts the unit cubes that fill a solid. Use **cubic units** (cm³, in³, ft³).
- **V = l × w × h**, or **V = B × h**, where *B* is the area of the base.

**WORKED EXAMPLE**

A box is 7 cm long, 3 cm wide, and 5 cm tall.

**Step 1** One layer of cubes: B = 7 × 3 = 21

**Step 2** Stack 5 layers: 21 × 5 = 105

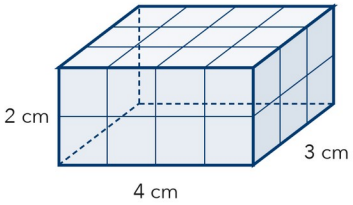
**Answer** **105 cm³**

## PRACTICE

*show your work · label units*

**1.** Each small cube is 1 cm³. Find the volume of the prism.

*Show your work.*




**Answer**_____ cm³

**2.** Find the volume of each rectangular prism.

**a.** 8 in × 5 in × 2 in _____ in³

**b.** 10 cm × 10 cm × 4 cm _____ cm³

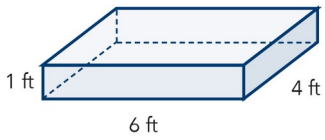
**c.** 6 ft × 6 ft × 6 ft _____ ft³

**d.** B = 45 m², h = 7 m _____ m³

**e.** 12 m × 3 m × 5 m _____ m³

**f.** B = 32 in², h = 9 in _____ in³

**3.** **Sandbox.** A sandbox is 6 ft long, 4 ft wide, and 1 ft deep. One bag of sand fills 2 ft³. How many bags fill the sandbox?




*Show your work.*

**Answer**_____ bags

**4.** **Fish Tank.** A fish tank is 20 in long and 10 in wide. It holds 3,000 in³ of water when full. How tall is the tank?

**Answer**_____ in

**5.** **Compare.** Box A is 6 in × 4 in × 2 in. Box B is 3 in × 4 in × 4 in. Mika says Box A holds more because it is longer. Is she right? Explain.

_____

_____

**CHECK YOURSELF** ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #___

**AT HOME** Find a box at home. Measure or estimate its length, width, and height, and find its volume.

Name_____ Date_____ Period_____

**LESSON 5-5 · HOMEWORK PACKET · SHEET 2 OF 2**

# Missing Fractional Edges

Maryland Standard 6.G.A.2 · Learning target – I can find a missing edge of a prism and find volumes with fractional edge lengths.

**REMEMBER**

- **V = l × w × h** works with fractions too. Write mixed numbers as fractions first.
- Missing edge: **divide** the volume by the product of the other two edges.

**WORKED EXAMPLE**

A box is 2 ½ ft × 2 ft × 1 ½ ft. Find its volume.

**Step 1** Write as fractions: 5/2 × 2/1 × 3/2

**Step 2** Multiply: (5 × 2 × 3)/(2 × 1 × 2) = 30/4

**Step 3** Simplify: 30/4 = 7 ½

**Answer** **7 ½ ft³**

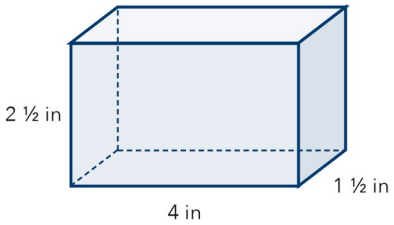
## PRACTICE

*show your work · label units*

**1.** Find the volume of the prism.

*Show your work.*

**Answer**_____ in³




**2.** Find each volume.

**a.** ½ ft × 6 ft × 8 ft _____ ft³

**b.** 3 ½ m × 2 m × 4 m _____ m³

**c.** 2 ½ cm × 4 cm × 3 cm _____ cm³

**d.** 1 ½ in × 1 ½ in × 4 in _____ in³

**e.** 4 ft × ½ ft × 5 ft _____ ft³

**f.** 6 cm × 1 ½ cm × 3 cm _____ cm³

**3.** **Planter.** A planter box is 3 ft long and 1 ½ ft wide. It holds 9 ft³ of soil when full. How deep is it?

**Answer**_____ ft

**4.** **Small Cubes.** A box is 2 in long, 1 in wide, and 1 ½ in tall. It is packed with cubes that are ½ in on each edge.

**a.** How many cubes fill the box? _____

**b.** Each cube is 1/8 in³. What is the volume of the box? _____ in³

## TEST PRACTICE

*choose the best answer*

**5.** A box is 5 in long, 2 in wide, and 1 ½ in tall. What is its volume?

**A** 8 ½ in³ **B** 10 in³ **C** 15 in² **D** 15 in³

**6.** A prism has a volume of 96 ft³. Its base is 4 ft by 6 ft. What is its height?

**A** 4 ft **B** 16 ft **C** 24 ft **D** 576 ft

**CHECK YOURSELF** ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #___

**AT HOME** Find a box at home with a fraction edge (like 2 ½ in). Estimate its volume.